LAB ASSIGNMENT – 6
DATE: 02/03/2026

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RegNO: AU/2023/0010993
Section :'E'

Schema

{ emp_id, name, department, designation, city, salary, experience }

Initial Data

Emp ID	Name	Department	Designation	City	Salary	Experience
101	Amit	IT	Developer	Kolkata	60000	3
102	Neha	HR	HR Executive	Delhi	45000	2
103	Ravi	IT	Tester	Mumbai	50000	4
104	Pooja	Finance	Accountant	Chennai	55000	5
105	Karan	IT	Developer	Kolkata	65000	6
106	Sneha	HR	Manager	Delhi	70000	7
107	Arjun	Finance	Analyst	Mumbai	48000	2
108	Meera	IT	Developer	Bangalore	75000	8
109	Rohit	IT	Team Lead	Pune	90000	10
110	Anjali	HR	Recruiter	Kolkata	42000	1

Lab Questions

Q1. Display employee name and salary only.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find({}, {name:1,salary:1,_id:0})
[
  { name: 'Amit', salary: 60000 },
  { name: 'Neha', salary: 45000 },
  { name: 'Ravi', salary: 50000 },
  { name: 'Pooja', salary: 55000 },
  { name: 'Karan', salary: 65000 },
  { name: 'Sneha', salary: 70000 },
  { name: 'Arjun', salary: 48000 },
  { name: 'Meera', salary: 75000 },
  { name: 'Rohit', salary: 90000 },
  { name: 'Anjali', salary: 42000 }
]
```

Q2. Find employees working in HR department.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find({department:"HR"})
[
  {
    _id: ObjectId('6992d717147b27b4c01e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6992d717147b27b4c01e262a'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

Q3. Find employees whose salary is between 50,000 and 80,000.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find({salary:{$gte:50000,$lte:80000}})
[
  {
    _id: ObjectId('6992d717147b27b4c01e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
```

```
  {
    _id: ObjectId('6992d717147b27b4c01e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  }
]
```

Q4. Display employees from Kolkata city.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find({city:"Kolkata"})
[
  {
    _id: ObjectId('6992d717147b27b4c01e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6992d717147b27b4c01e262a'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

Q5. Sort employees by experience in ascending order.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find().sort({experience:1})
[
  {
    _id: ObjectId('6992d717147b27b4c01e262a'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
```

```
  {
    _id: ObjectId('6992d717147b27b4c01e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
```

```
  {
    _id: ObjectId('6992d717147b27b4c01e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]
```

Q6. Display top 5 most experienced employees.

```
  {
    _id: ObjectId('6992d717147b27b4c01e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]
```

Q7. Update salary of employee with emp_id 102 to 48,000.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.updateOne({emp_id:102},{$set:{salary:48000}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q8. Increase salary of all IT employees by 10,000.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.updateMany({department:"IT"},{$inc:{salary:10000}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

Q9. Update designation to “Senior Developer” for IT employees having experience more than 5 years.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.updateMany({department:"IT",experience:{$gt:5}},{$set:{designation:"Senior Developer"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

Q10. Delete employees whose experience is less than 2 years.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.deleteMany({experience:{$lt:2}})
{ acknowledged: true, deletedCount: 1 }
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

Q11. Display employees who are not from Delhi.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find({city:{$ne:"Delhi"}})
[ acknowledged: true, deletedCount: 1 }
  {cknowledged: true,
    _id: ObjectId('6992d717147b27b4c01e2621'),
    emp_id: 101,3,
    name: 'Amit',3,
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 60000,
    experience: 4
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 75000,
    experience: 6
  },
```

```
  {
    _id: ObjectId('6992d717147b27b4c01e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Bangalore',
    salary: 85000,
    experience: 8
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Pune',
    salary: 100000,
    experience: 10
  }
]
```

Q12. Find employees whose designation is Developer and salary greater than 60,000.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find({designation:"Developer",salary:{$gt:60000}})
[
  {
    _id: ObjectId('6992d717147b27b4c01e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  }
]
```

Q13. Display employees working in IT or Finance department.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find({department:{$in:["IT","Finance"]}})
[
  {
    _id: ObjectId('6992d717147b27b4c01e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 60000,
    experience: 4
```


```
    salary: 60000,
    experience: 4
```


```
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 75000,
    experience: 6
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Bangalore',
    salary: 85000,
    experience: 8
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Pune',
    salary: 100000,
    experience: 10
  }
]
```

Q14. Display employees sorted by department and then by salary.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find().sort({department:1,salary:1})
[
  {
    _id: ObjectId('6992d717147b27b4c01e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
```

```
  {
    _id: ObjectId('6992d717147b27b4c01e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 75000,
    experience: 6
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Bangalore',
    salary: 85000,
    experience: 8
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Pune',
    salary: 100000,
    experience: 10
  }
]
```

Q15. Count total number of employees in the company.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.countDocuments()
9
```

Q16. Display employees whose salary is not equal to 50,000.

```
EmployeeDB_AIML075> db.EmployeeList_AIML075.find({salary:{$ne:50000}})
[
  {
    _id: ObjectId('6992d717147b27b4c01e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 60000,
    experience: 4
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
```

```
    experience: 6
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Bangalore',
    salary: 85000,
    experience: 8
  },
  {
    _id: ObjectId('6992d717147b27b4c01e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Pune',
    salary: 100000,
    experience: 10
  }
]
```